

Multiple Choice (10 marks)

1. The symbol for antimony is
 - (a) Am
 - (b) As
 - (c) Ab
 - (d) Au
 - (e) Fe
2. A saturated solution of CaF_2 contains .00168 g of CaF_2 per 100 g of water. Determine the K_{sp} .
 - (a) 2.11×10^{-8}
 - (a) 2.73×10^{-8}
 - (a) 1.08×10^{-8}
 - (a) 3.92×10^{-8}
 - (a) 4.77×10^{-8}
3. Write the orbitals, in the order of increasing energy, for hydrogen - like atoms.
 - (a) $1s < 2p < 2s < 3p < 3s < 3d$
 - (b) $2s < 2p < 1s < 3s < 3p < 3d$
 - (c) $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < 5s = 5p = 5d = 5f = 5g$
 - (d) $2s = 2p < 1s = 3p = 3d$
 - (e) $1s < 3s < 2s < 5s < 4s$
4. Use the van der Waals parameters for chlorine to calculate approximate values of the Boyle temperature of chlorine.
 - (a) $1.25 \times 10^3 \text{ K}$
 - (b) $1.20 \times 10^3 \text{ K}$
 - (c) $1.41 \times 10^3 \text{ K}$
 - (d) $1.10 \times 10^3 \text{ K}$
 - (e) $1.65 \times 10^3 \text{ K}$

5. The +1 oxidation state is more stable than the +3 oxidation state for which group 13 element?
- (a) Al
 - (b) Tl
 - (c) C
 - (d) B
 - (e) Si

True / False (10 marks)

Answer True or False

- 6. True or False: The solutions for the equation $x^2 - 5x + 6 = 0$ are $x = 2$ and $x = 3$.
- 7. True or False: CH_3NH_2 can form hydrogen bonds due to the presence of a nitrogen atom bonded to hydrogen.
- 8. True or False: The ^{13}C NMR spectrum of 3-ethylpentane exhibits five distinct chemical shifts due to its unique structural environment.
- 9. True or False: High melting point substances that do not dissolve in water typically exhibit ionic bonding characteristics.
- 10. True or False: The logarithm of 3^2 is equal to 6, as it can be expressed as 2 times the logarithm of 3.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the factors that influence the chemical shift in NMR spectroscopy, and analyze why CH_3Cl exhibits a ^1H resonance approximately 1.55 kHz from TMS in a 12.0 T magnetic field.
- 12. Discuss the process by which Cobalt-60 is produced through the bombardment of Cobalt-59. Analyze its significance in radiation therapy for cancer treatment.

End of Paper